KITCHENERY
FREEDOM TO COOK

KIT_QEP_TX3.0 AC-DC Module. Electronic circuit specification and requirements

File name	**KIT_QEP_TX3.0 AC-DC Module. Electronic circuit specification and requirements**
Date	July 22, 2024
Author	Akshay Bhuva
Description	Product specification and requirements for the development of the electronic circuits inside the **KIT_QEP_TX3.0** Module

Version control record

Issue	Description of changes	Date	Author
01	Initial version	07/22/2024	Akshay Bhuva, Daniel Loranger

© 2024 Kitchenery Inc. Proprietary & Confidential

Table of contents

KIT_QEP_TX3.0 AC-DC Module. Electronic circuit specification and requirements 1

1. Introduction of Kitchenery 3

2. Purpose of this project 3

3. Scope 4

4. KIT_QEP_TX3.0 Module description and technical requirements 5

4.1. EMI Filter circuitry 5

4.2. Combined EMI + AC/DC PCB 6

4.2.1. Input 7

4.2.2. AC/DC Converter Module 7

4.2.3. Output current sensor 7

4.2.4. Power output connector 8

4.2.5. Interconnecting wires requirements 8

5. PCB Requirements 9

6. Project Management Requirement 10

6.1. Milestones and budget 10

6.2. List of deliverables requirements 11

6.3. Design process requirements 12

6.4. Delivery of goods requirements 13

List of figures

Figure 1: KIT_QEP_TX3.0 architecture. 5

List of tables

Table 1: List of minimum deliverables. 11

© 2024 Kitchenery Inc. Proprietary & Confidential

1. Introduction of Kitchenery

Kitchenery is a technological start-up company devoted to providing wireless power transfer technology, smart connectivity, and IoT solutions for small kitchen appliances. The company is focused on the development and production of wireless technology products that make life easier. No more messy wires, overloaded sockets, and electrocution hazards.

With Kitchenery's modular & innovative wireless power transfer system, you can have the world's best cooking experience that will set you free in your kitchen.

- **Product Overview:** Our system consists of a power transmission and reception module, all of them with a potential of smart IoT integration, this is facilitated by the presence of Wi-Fi and Bluetooth capabilities integrated with-in the system.
    - **QUANTUM ENERGY PAD (QEP):** This is a dual-function transmitter, which can be used as an induction base + wireless power transmitter module. The wireless power transmission function is integrated with the induction heating function. The hybrid transmitter adopts magnetic resonance technology and an intelligent control system, which can automatically switch the power mode between wireless power transmission and induction heating. This transmitter can be sold in 2 different variety of product categories:
        - A countertop built-in or portable hybrid induction cooktop (2 or 4 burner top)
        - Installed underneath the kitchen countertop, under the kitchen furniture (such as the dining table), or even integrated with the kitchen countertop. It works with any non-metal countertop (Stone, marble, wood, granite, soapstone, quartz, resin, laminate, concrete, etc.). This module is ideally hidden giving a truly aesthetically pleasing kitchen experience.
        - **This project involves designing an AC-DC power supply board for the QEP.**

- **Wireless power receiver module:** This module is integrated with kitchen appliances such as juicer, toaster, air fryer, blender and so on. When this module is placed on a transmitter, it can start receiving wireless power.

Our patent-pending technology has been already evaluated in a proof of concept and now we are ready to take new steps to have our first commercial ready-to-sell product.

2. Purpose of this project

Kitchenery wants to build a new device with improved performances and to achieve that, it is necessary to develop new electronic circuitry for our KIT_QEP_TX3.0A Module. We intend to ask for a quotation to outsource the design and interconnection harnesses of this new electronic circuits. The purpose of this document is to describe the architecture of the KIT_QEP_TX3.0A AC-DC MODULE and to set the main requirements in order to be used as an input to perform the

© 2024 Kitchenery Inc. Proprietary & Confidential

design, manufacturing and testing of the electronic circuits in this module. This document contains the following information:

- Scope of the activities that will be outsourced.
- A description of the architecture of the module and a list of requirements.
- A list of deliverables.
- Project management, schedule and milestones requirements.

3. Scope

In order to build our new V3 TX module, Kitchenery need to develop new AC-DC electronic board to power all the subsystems within the QEP . We are seeking an experienced Schematic Designer / PCB Designer to develop an AC/DC board that will serve as a 1.5kW DC power supply with multiple outputs: 1) High voltage 100Vdc output at ~16-18A (1.5kW for the coil), and several smaller outputs. The design must be robust, meet all UL safety requirements, and include features for short circuit output protection and regulation within 10% from 5W to 1500W loads (100Vdc output). The final design should be open frame with a maximum height of 25mm.

The scope of the activities to be outsource in the framework of this project are:

- Design of an AC-DC PCB.
    - Develop a reliable and efficient 1.5KW AC/DC power supply, with several low-voltage outputs.
    - Ensure the design meets all relevant safety standards and regulations.
    - Achieve a compact design with a maximum height of 25mm or lower.
    - Provide a robust solution that includes necessary protection features and stable output regulation.
    - Achieve cost efficient BOM solution for consumer electronics.
- Manufacturing, assembly and testing of 2 (two) AC-DC PCBs

© 2024 Kitchenery Inc. Proprietary & Confidential

4. KIT_QEP_TX3.0 Module description and technical requirements

This section describes the architecture of the KIT_QEP_TX3.0 module and the high-level technical requirements to develop these PCBs and cable harnesses. The main architecture of this module is shown in the following figure. The KIT_QEP_TX3.0 AC-DC Module is a part of the main architecture.

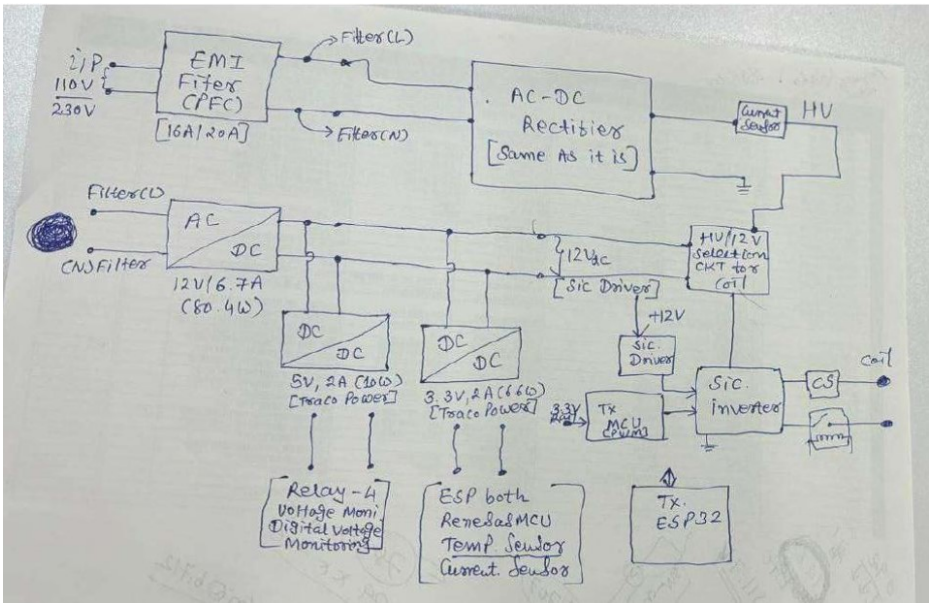


Figure 1: KIT_QEP_TX3.0 architecture.

Note: the above diagram is missing the 24V output for the AC-DC driver.

As it is shown in the figure above, the main, and only electrical input of the system is the AC input from mains. This architecture is not a requirement, its one potential solution. If you have a better architecture, feel free to discuss with us.

REQ_FUN_1. The system shall be able to work using the most common power input, that is, 230VAC 50Hz and 115VAC 60Hz. Including all transient spikes (typical power line problems).

REQ_FUN_2. EMI filter PCB shall be able to provide power to the AC-DC power conversion PCB. The design shall include all EMI/ EMC mitigations.

In the next sections, each one of the following main parts of the module will be described.

4.1. EMI Filter circuitry

The purpose of this circuitry is to filter the conducted electromagnetic emissions from the module in order to comply with the standard certification such as CE, UL and FCC mark. It must ensure Class-B compliance for home use.

© 2024 Kitchenery Inc. Proprietary & Confidential

REQ_FUN_3. The EMI filter shall be a passive filter based on capacitors, coils, chokes and similar components.
REQ_FUN_4. The EMI Filter shall include a fuse holder to place a fuse to protect the module.
REQ_FUN_5. Maximum power consumption of the module is 3.5kW, therefore, maximum current in the EMI Filter circuitry shall be higher than 32A(*) when working at 115VAC. Maximum power consumption in each main control PCB is 1.8kW and therefore maximum current in the outputs is 16A(*).
REQ_FUN_6. EMI Filter shall have a terminal for earthing purpose.
REQ_FUN_7. The board shall have 2 input terminals to connect AC input. These terminals shall allow to connect an O-ring terminal for screws M5. These terminals shall withstand a 32A(*) current.
REQ_FUN_8. Maximum attenuation of this filter PCB shall be in the range between 10kHz to 80kHz.
REQ_FUN_9. All fixation holes shall have a copper plated area the allow to connect the fixation screws to the metallic enclosure of the module.

(*) Note: current values for AC signals are RMS value, not maximum amplitude current. This will be true henceforward in this document.

4.2. Combined EMI + AC/DC PCB

This PCB is made up of the following blocks:

- EMI Circuitry.
- AC/DC converter. This module converts the input AC voltage to the necessary voltage levels for the control electronics. The outputs will be isolated from the AC voltage. The output voltage levels of this converter will be at least:
    - High Voltage Vdc output = 100Vdc at approx. 16-18Amps for 1.5kW.
    - 3.3Vdc at approx. 4amps.
    - 5Vdc at approx. 2amps.
    - 12Vdc at approx. 2amps.
    - 24Vdc at approx. 2amps.

- Must include a power management IC with voltage and current monitoring for all outputs.
- All control signals are optically isolated
- External connector. This board has the following external connectors:
    - Power output connector, we need the high voltage separate from the low voltage
    - Ganged connectors for all other low power outputs.

REQ_FUN_10. Maximum size of the AC/DC PCB will be 10cm x 10cm.
REQ_FUN_11. Maximum heigh of the PCB will be less than 2.5cm. preference should be given to shorter components where possible.

© 2024 Kitchenery Inc. Proprietary & Confidential

REQ_FUN_12. AC/DC PCB will have 4/ TBD holes to assembly the PCB. These holes will allow M3 screws to be used for this purpose. Position and size of these wholes will be agreed during the design stage. These holes shall have a copper plated area.

REQ_FUN_13. Since this PCB shall be assembly in an enclosure, the total height, position of the components and 3D design shall be agreed between Kitchenery and the supplier.

In the following sections, the requirements of each one of the modules in the Main Control Board is listed.

4.2.1. Input

REQ_FUN_14. One side of the board is the Vac input. The input connector will consist of two male 250" FASTON terminals, capable to withstand 20amps.

REQ_FUN_15. Interface to the processor, control connectivity located in the low voltage DC area. Connector TBD.

REQ_FUN_16. Just after the input connector, the AC input shall have safety fusing. Fuse specs TBD.

REQ_FUN_17. Module shall include soft start feature, to overcome any kind of huge current spikes.

4.2.2. AC/DC Converter Module

REQ_FUN_18. The AC/DC converter outputs will be isolated from the AC input using a transformer. The use of either switched power supplies or linear regulators is up to the supplier provided that the solution meets all the requirements.

REQ_FUN_19. The AC/DC converter module will have 1 high voltage output and 4 low voltage outputs.

REQ_FUN_20. First output will be a 100Vdc with a maximum output power of 1500W or higher.

REQ_FUN_21. Second output will be 3.3Vdc, 4A

REQ_FUN_22. Third output will be 5Vdc, 2A

REQ_FUN_23. Fourth output will be 12Vdc, 2A

REQ_FUN_24. Fifth output is 24Vdc, 2A

REQ_FUN_25. The high voltage and low voltage systems must be mechanically isolated connectors (separate connectors)

REQ_FUN_26. Type of connectors - TBD

4.2.3. Output current sensor

REQ_FUN_27. Must include a power management IC with voltage and current monitoring for all outputs.

REQ_FUN_28. For high voltage output, sensor shall have a measurement range between 0 and 40A.

REQ_FUN_29. Resolution of the measurement shall be at least 0.1A.

© 2024 Kitchenery Inc. Proprietary & Confidential

REQ_FUN_30. Accuracy of the measurement shall be 1%FSR.
REQ_FUN_31. Current measurement shall include a low pass filter with a cutoff frequency of 500kHz (TBD).

4.2.4. Power output connector

REQ_FUN_32. This connector will connect the outputs of the AC-DC boards to various systems of the. Output connectors – TBD.

4.2.5. Interconnecting wires requirements

REQ_FUN_33. Type and section of interconnecting wires will be UL approved materials.
REQ_FUN_34. Section of the wire shall be enough to drive the expected current at each interface according to US National Electrical Code (NEC) or equivalent recognized standard.
REQ_FUN_35. Wires shall be PVC-insulated.
REQ_FUN_36. Connectors of interconnecting wires shall be according to the requirements of the different PCBs.

© 2024 Kitchenery Inc. Proprietary & Confidential

5. PCB Requirements

REQ_FUN_37. PCB shall have 4 layers maximum. If more than 4 will be needed, this should be approved by Kitchenery.
REQ_FUN_38. Total PCB thickness shall be 1.6mm.
REQ_FUN_39. PCB shall include a green solder mask.
REQ_FUN_40. PCB shall include a white silkscreen with at least the designator and the shape of the component.
REQ_FUN_41. PCB shall be IPC-A-600 class 2 or equivalent.
REQ_FUN_42. PCB surface finish shall be immersion silver or ENIG.
REQ_FUN_43. Track width and cooper thickness shall be calculated according to IPC-2221 or similar international standard. To calculate the size, a maximum temperature rise of 20 degrees shall be considered.
REQ_FUN_44. Design must strive for off-the-shelf heat sink solutions
REQ_FUN_45. Layout should follow IPC 2221A and IPC 9592 standards, UL 60950-1
REQ_FUN_46. Have an Independent Design Review of the schematics and layout from a knowledgeable 3rd party supplier.

© 2024 Kitchenery Inc. Proprietary & Confidential

6. Project Management Requirement

This section describes a set of requirements that shall be considered for the development of the KIT_QEP_TX3.0 AC-DC Module. The following are the agreements that must be signed with the supplier:

- Kitchenery NDA
- Kitchenery Contractor and SOW agreement

Among other this section describes the deliverables, the main milestones and reviews and other management-related requirements.

6.1. Milestones and budget

REQ_MNG_1. During the development of the KIT_QEP_TX3.0 AC-DC module the following formal reviews will be performed between Kitchenery and the supplier.

- Kick-off meeting. During this meeting this specification will be reviewed, and the supplier may raise questions needed to understand the requirements. Supplier will present at least a project schedule. Moreover, a final list of deliverables and a preliminary delivery date for each one of them will be agreed.
- Preliminary Design Review (PDR). During this milestone the supplier shall deliver a block diagram of the final solution together with a selection of the main components of the product.
- Critical Design Review (CDR). During this milestone, the supplier will deliver the low-level design, schematic, PCB layout, preliminary test procedure, etc. The design will be reviewed by Kitchenery and some changes may be requested. Once the design is approved, the PCBs will be ready to be manufactured and tested.
- Preliminary Acceptance (PA). Once the PCBs has been tested at supplier's site, the results of the test campaign will be reviewed. Once results have been accepted by Kitchenery, they will be sent to our facilities to be tested.
- Final Acceptance (FA). PCBs will be tested by Kitchenery prior to have the final acceptance and therefore of considering the project as closed.

REQ_MNG_2. The budget shall be divided into the above milestones. The amount to be invoice in each one of the milestones shall be proposed by the supplier and finally agreed by both parties.

REQ_MNG_3. Budget shall be divided into, at least, the following concepts:

- Engineering costs.
- Testing costs.
- Materials costs.
- Shipment costs.
- Travel expenses (if necessary).

© 2024 Kitchenery Inc. Proprietary & Confidential

6.2. List of deliverables requirements

REQ_MNG_4. The following table contain a list of minimum deliverables during the development of the KIT_HYB_WPT_TX1.0A module.

#	Deliverable name	Type	Expected delivery
1	Technical offer	Document	Call for Tender
2	Commercial offer	Document	Call for Tender
3	Preliminary schedule	Document	Call for Tender
4	Project Schedule	Document	KoM
5	High level design (architecture + main components)	Document	PDR
6	Design documents (Project file, detailed schematics, PCB Layout, Libraries, etc.)	CAD files	CDR
7	Design document	Document	CDR
8	3D Model	CAD files	CDR
9	Test case definition	Document	CDR
10	Traceability matrix	Document	CDR
11	Complete bill of materials	Document	CDR
12	AC-DC Boards	Product	PA
13	Cable harnesses	Product	PA
14	Test reports for every product delivered	Product	PA

Table 1: List of minimum deliverables.

- **Schematic Design**:
  - Detailed schematic of the AC/DC power supply.
  - Block diagram showcasing the overall design and component layout.
  - Component selection and justification for critical components.
- **PCB Design**:
  - Layout design that fits the specified dimensions (maximum height of 25mm).
  - Placement of components ensuring optimal performance and safety.
  - Design files compatible with standard PCB manufacturing processes.
- **Safety and Compliance**:
  - Ensure design meets UL safety requirements.
  - Incorporate short circuit output protection.
  - Regulation to within 10% from 5W to 1500W loads.
- **Documentation**:
  - Bill of Materials (BOM) with part numbers and specifications.
  - Design and testing documentation.
  - Compliance reports and safety analysis.

REQ_MNG_5. All deliverables shall be written in English language.

REQ_MNG_6. All design documents will be delivered to Kitchenery including CAD schematic files, CAD PCB design, libraries, 3D models, etc.

© 2024 Kitchenery Inc. Proprietary & Confidential

REQ_MNG_7. A 3D model of the PCB will be delivered. For those components for which a 3D model can be found on the Internet, using a simplified body such as a cylinder or a cube will be accepted.
REQ_MNG_8. Electronic design shall be done using Altium Software.
REQ_MNG_9. Spice simulations shall be made using Altium Software.
REQ_MNG_10. Resistors in schematic file shall contain a field with the following information: Footprint (0603, 0805, etc. or AXIAL), rated power and tolerance. Example 0603 60mW 1%. This field shall be visible on the sheet.
REQ_MNG_11. Capacitors in schematic file shall contain a field with the following information: footprint/size (0603, 0805, 15diam, etc.), rated voltage and tolerance.
REQ_MNG_12. Every polarized capacitor shall be identified in the schematic.
REQ_MNG_13. Inductors in schematic file shall contain a field with the following information: footprint/size (0805, AXIAL, etc.) rated current and tolerance.
REQ_MNG_14. Resistors, capacitors and inductors in schematic shall contain a field called value with the resistance, capacitance or inductance value.
REQ_MNG_15. Diodes shall have a field visible on the schematic with the following information: rated current, reverse voltage.
REQ_MNG_16. Transistor shall have a field visible on the schematic file with the following information: Type (NPN, nMOS, etc.). maximum reverse voltage and maximum current.
REQ_MNG_17. Every other component (Op Ams, comparators, ADC, etc.) shall have a visible field with the product name (for instance INA122, NCP51530, etc.).
REQ_MNG_18. Every component in the schematic file shall contain a hidden field called P/N indicating the full part number of the component. This may only be skipped for resistors, capacitors and coils.
REQ_MNG_19. Whenever possible, footprints and schematic components from Altium standard library shall be used. When not possible, the schematic symbol and/or footprint will be created. The supplier is allowed to use their own libraries when the component/footprint does not exist in the standard libraries. However, these symbols shall be delivered to Kitchenery.
REQ_MNG_20. Traceability matrix shall trace all functional requirements with a design element in the circuit and also with the mean to guarantee the compliance either a test case in the test case procedure document or a document including an analysis that guarantee the requirement.

6.3. Design process requirements

REQ_MNG_21. For thermal analysis to calculate heatsinks a safety margin of a 15% shall be considered and maximum junction temperature shall be kept below 70ºC.
REQ_MNG_22. Supplier is encouraged to propose modifications, to comply partially or not comply with some of the requirements if that means either a significant improvement in performances of the systems or a significant reduction in costs/lead time. However, any modification to this requirement definition

© 2024 Kitchenery Inc. Proprietary & Confidential

document shall be previously agreed and approved by Kitchenery. This analysis, especially those change proposals affecting the cost of the development, should be made during the tender process in order to get the best value for money.

REQ_MNG_23. After CDR is closed, supplier shall inform to Kitchenery about all the problems found during the tender stage using a problem report/change proposal form. This document shall describe the problem, the components affected by the problem, the changes necessary in the design (if any). Also, after a problem is resolved, design document shall be updated to a different version.

REQ_MNG_24. Supplier shall deliver fully functional PCBs that comply with the requirements defined in this document or those requirements agreed during the development. If there are design errors and it is necessary to manufacture several batches, this will be borne by the supplier. It's supplier responsibility to propose the best plan to reduce risks and avoiding extra costs.

6.4. Delivery of goods requirements

REQ_MNG_25. All material goods, PCBs, spare materials (if any), cable harnesses, etc., shall be sent to DAP (Delivery at Place) to U.S.

© 2024 Kitchenery Inc. Proprietary & Confidential